Branching and Merging Code with Git Source Control SOP

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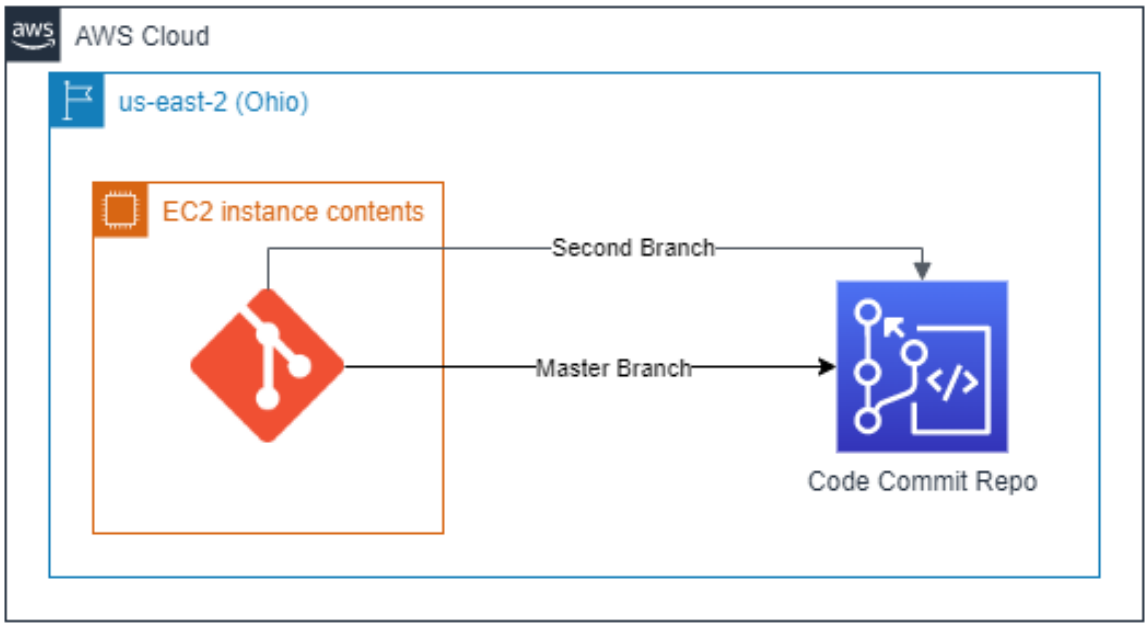
1. Branching and Merging Code with Git Source Control

1.1 Description

When using Git for source control and versioning, the ability to branch a repository while you make changes off the mainline is key. However, eventually, that branch will need to re-join the master branch and you will need to perform a merge. This is how a single Code Base is used to perform multiple developmental activities.

1.2 Architecture Diagram

The diagram below displays a visual representation of the application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Create a CodeCommit Repository.
2. Create an EC2 Instance (Amazon Linux 2) and SSH into it using the Sessions Manager Option.
   Note: Make sure **CCL-EC2-Role** is attached to the instance, only then SSH is possible with Sessions Manager Option.
   a. Use the command to install git:

   ```
   sudo su
   sudo yum install git -y
   ```

   b. Install AWS CLI (Refer the third link in Supporting References section).
   c. Use **aws configure** after installing and set region to **us-east-2**. Press enter for rest of the options.
   d. Install Unzip (install if needed).
3. Configure the system with the user email and user name:

   ```
   git config --global user.email <some sample mail ID>
   git config --global user.name <some name>
   ```

4. Navigate to **IAM** > **Users** > **Your UserName** > **Security Credentials** > **HTTPS Git Credentials** for CodeCommit. Here generate credentials and save them.
5. Clone the CodeCommit repository using the git clone command which can be found in Step 3 under Connection Steps in created repository.
6. Go to the repo folder:

   ```
   cd <destination path>
   ```

7. Create some text file with some content in the folder.
8. Now give the following commands to push the file to the repository:

   ```
   git add <filename>
   ```

   ```
   git commit -m "Some Commit Message for reference"
   ```

   ```
   git push
   ```

   Note: The git push command pushes the code by default to Master Branch.
9. Verify if the file is reflected in the **Code commit** Console Page.
10. Create a new branch:

    ```
    git branch <some branch name>
    ```

11. Switch to new branch with command:

    ```
    sudo git checkout <branch-name>
    ```

12. Create some file in the newly created branch.
13. Using the following command, find the untracked files and changes made in the files.

    ```
    git status
    ```

14. Commit all the changes:

```
git add <filename>
```

```
git commit -m "<Some Message>"
```

```
git push --set-upstream origin <branchname>
```

15. Check in CodeCommit if the newly created branch is available there.
16. Switch to master branch from current branch using the following command:

```
checkout
```

17. Execute the following command to merge the branches:

```
aws codecommit merge-branches-by-fast-forward --source-commit-specifier <branch-name> --destination-commit-specifier master --repository-name <repo-name>
```

18. Check the CodeCommit page to see if both the files are available in the master.

1.4 Troubleshooting

S. No	Problem	Solution
1	Git command not found in your terminal	Make sure git is installed: **`sudo yum install git`**
2	Branch not found	Give branch name correctly while checkout to that branch.

1.5 Supporting References

Refer the below links for additional information:

1. https://git-scm.com/book/en/v2/Git-Branching-Basic-Branching-and-Merging
2. https://www.atlassian.com/git/tutorials/using-branches
3. https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-linux.html